SUBSCRIPTION BUSINESS ANALYTICS: REVENUE, CHURN & CUSTOMER INSIGHTS DASHBOARD

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Problem Statement

Subscription-based businesses often struggle to understand key drivers of revenue growth, customer retention, and churn. Without clear visibility into these metrics, it becomes difficult to identify revenue opportunities, reduce customer loss, and optimise pricing strategies.

This project delivers an end-to-end analytics solution to track customer behaviour, revenue performance, and churn trends, enabling data-driven decision-making.

Dataset Overview

The dataset consists of multiple tables representing a subscription-based business model:

- Customers: Customer demographics, country, and acquisition details
- Subscriptions: Customer subscription lifecycle (start date, end date, plan)
- Payments: Transaction-level data including payment status and amount
- Invoices: Billing records and invoice status
- Subscription Plans: Plan types, pricing, and billing cycles
- Churn Events: Records of customer churn occurrences
- Plan Changes: Upgrade and downgrade tracking

A date table was also created to support time-based analysis.

Methodology

A structured data modelling approach was used to connect all tables through CustomerID and Date relationships, ensuring accurate filtering and aggregation across the dataset.

Key steps included:

- Data cleaning and transformation using Power Query
- Creating a star-schema model for efficient querying
- Developing DAX measures for KPIs such as:
    - Monthly Recurring Revenue (MRR)
    - Annual Recurring Revenue (ARR)
    - Average Revenue Per Customer (ARPU)
    - Churn Rate
- Building interactive dashboards to analyse trends over time, customer behaviour, and geographic performance

Dashboard Overview

The solution consists of four main dashboards:

1. Executive Overview
    - High-level KPIs (MRR, ARR, ARPU, Total Revenue)
    - Revenue growth trends
    - Payment performance

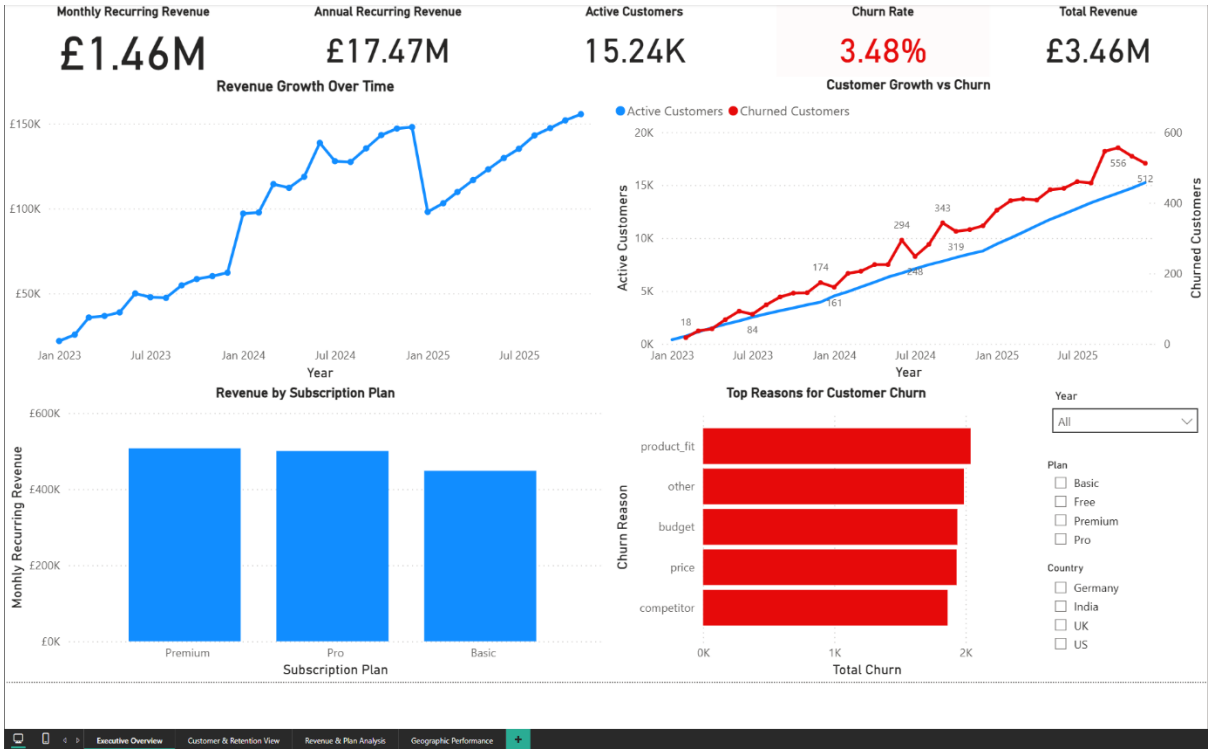

Figure 1: Executive Overview Dashboard

2. Customer & Retention View
    - Customer acquisition trends
    - Active vs churned customers
    - Churn rate analysis

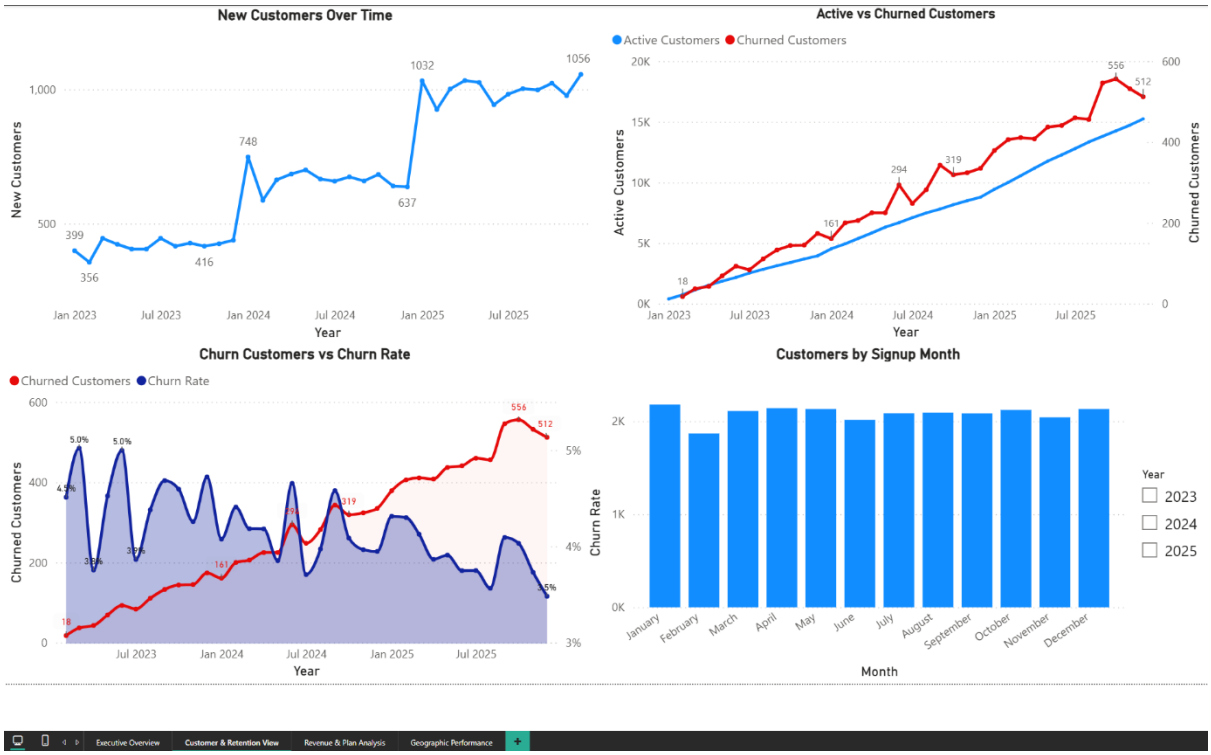

Figure 2: Customer & Retention Dashboard

3. Revenue & Plan Analysis
    - Revenue by subscription plan
    - Upgrade trends
    - Average revenue per customer over time

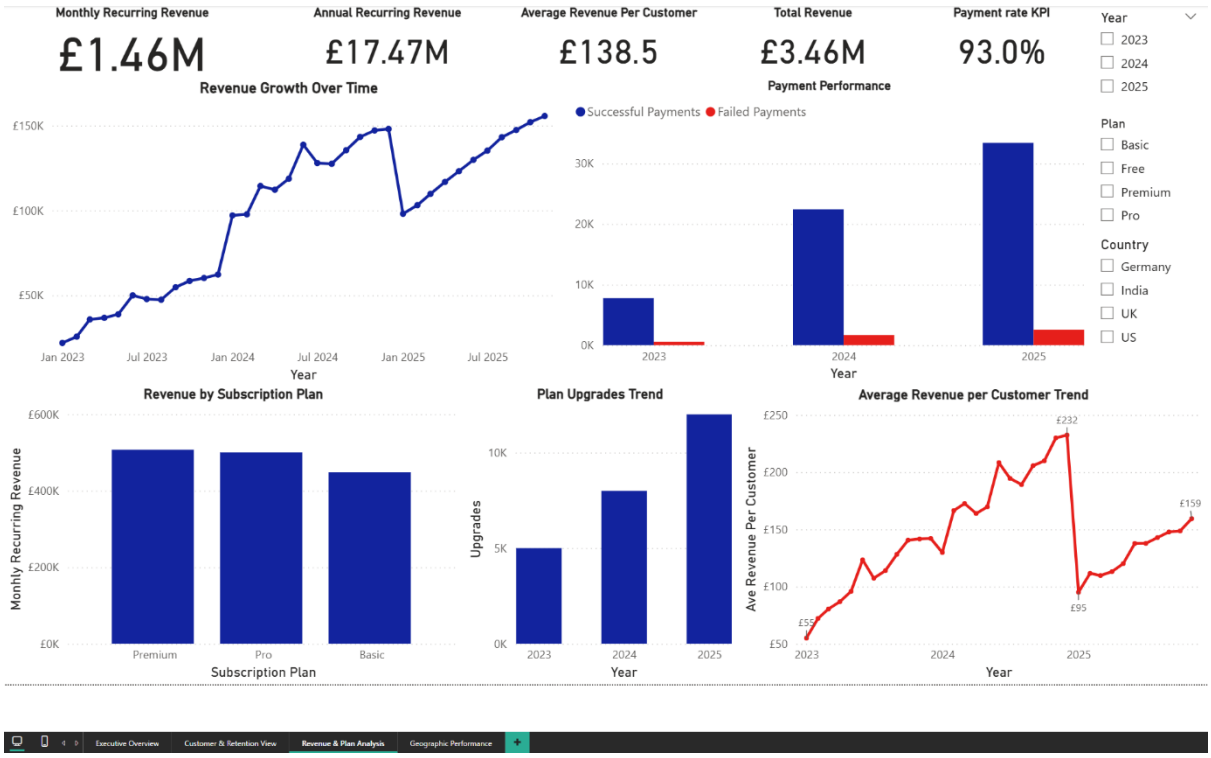

Figure 3: Revenue & Plan Analysis Dashboard

4. Geographic Performance
    - Revenue by country
    - ARPU by country
    - Customer distribution
    - Churn rate by country

Figure 4: Geographic Performance Dashboard

Key Insights

- Revenue has shown consistent growth over time, indicating strong business expansion.
- The UK generates the highest average revenue per customer, suggesting stronger monetisation compared to other regions.
- Customer distribution is relatively even across all countries, meaning no single market dominates the business.
- Churn rates remain consistently high across all regions (~38–40%), indicating a systemic retention issue rather than a location-specific problem.
- Subscription upgrades have increased over time, contributing positively to revenue growth.
- Payment success rates are high overall, but failed payments still represent a potential revenue leakage point.

Business Recommendations

- Focus on improving customer retention strategies, as high churn is consistent across all regions.
- Investigate customer experience, pricing, or product value to identify root causes of churn.
- Leverage the UK market's higher ARPU by analysing successful monetisation strategies and replicating them in other regions.
- Implement proactive churn prevention strategies such as:
    - Customer engagement campaigns
    - Loyalty incentives
    - Early churn detection models
- Reduce failed payments by introducing retry mechanisms or payment reminders.
- Continue encouraging subscription upgrades through targeted upselling strategies.

Tools Used

- Power BI (Data modelling, DAX, dashboard development)
- Power Query (Data cleaning and transformation)
- DAX (KPI calculations and measures)

Conclusion

This project demonstrates how data analytics can be used to uncover key business insights in a subscription-based model. By combining data modelling, DAX calculations, and interactive dashboards, the solution provides a clear view of revenue performance, customer behaviour, and retention challenges.

The insights generated can support strategic decisions aimed at increasing revenue, improving customer retention, and optimising overall business performance.

This solution can be directly applied to subscription-based businesses to identify revenue opportunities, reduce churn, and support data-driven decision-making.